Research in International Business and Finance 64 (2023) 101872

Contents lists available at ScienceDirect

Research in International Business and Finance

journal homepage: www.elsevier.com/locate/ribaf

Full length article

Does high-frequency trading actually improve market liquidity? A comparative study for selected models and measures

Renata Karkowska [a,*], Andrzej Palczewski [b]

[a] Faculty of Management, University of Warsaw, Szturmowa Street 1/3, 02-678 Warsaw, Poland
[b] Faculty of Mathematics, Informatics, and Mechanics, University of Warsaw, Banach Street 2, 02-097 Warsaw, Poland

ARTICLE INFO

JEL classification:
G10
G12
G14
G23

Keywords:
High-frequency trading
Liquidity
Algorithmic trading
Market microstructure

ABSTRACT

The increasing volume of messages sent to the exchange by algorithmic traders stimulates a fierce debate among academics and practitioners on the impacts of high-frequency trading (HFT) on capital markets. By comparing a variety of regression models that associate various measures of market liquidity with measures of high-frequency activity on the same dataset, we find that for some models the increase in high-frequency activity improves market liquidity, but for others, we get the opposite effect. We indicate that this ambiguity does not depend only on the stock market or the data period, but also on the used HFT measure: the increase of high-frequency orders leads to lower market liquidity whereas the increase in high-frequency trades improves liquidity. We hypothesize that the observed decrease in market liquidity associated with an increasing level of high-frequency orders is caused by a rise in quote volatility.

1. Introduction

Technological innovations and growing competition have enormously increased the use of automated trading practices, reducing the participation of traditional human traders. This phenomenon has been present on stock exchanges for more than a decade and is constantly growing. Improvements in technology have enabled computers to make trading decisions 100 times faster than human traders (Hasbrouck and Saar, 2013). Most trading decisions in the financial markets are now executed in the "millisecond environment". The statistical data provided by Hendershott and Riordan (2013) highlights that algorithmic trading (AT) generated 52% of market order volume and 64% of limit order volume on the Deutsche Börse in 2008. Brogaard (2010) and Brogaard et al. (2014) report that for the years 2008–2009, HFT was responsible for 68.5% of dollar volume and participated in 74% of trades on NASDAQ. Hagströmer and Nordén (2013) reveal that HFT in NASDAQ-OMX Stockholm constituted the lion's share of trading volume (63%–72%) and limit order traffic (81%–86%) in the years 2011–12. Anagnostidis and Fontaine (2020) analyzing stocks from the CAC 40 index in 2015 estimate that high-frequency traders were responsible for 98.5% of orders, 99.5% of cancellations, and 89.5% of marketable orders. Algorithmic trading also accounts for a significant fraction of trading volume in emerging markets, as is documented by Dubey et al. (2017) who show that for stocks in the NIFTY 50 index of the Indian Stock Market in August 2013 96% of orders and 75% of trades were executed by algorithmic traders.

The short review in the preceding paragraph reveals that different names are used to describe computer-based trading. In some cases, the name used by the author(s) is implied by the identification flag the exchange is attaching to orders and/or trades (AT or HFT flag). But in many cases, the authors use proxy metrics which correlate with automated trading calling such a metric AT measure or HFT measure without paying enough attention to the distinction between the two names and even using the names AT

* Corresponding author.
E-mail addresses: rkarkowska@wz.uw.edu.pl (R. Karkowska), a.palczewski@mimuw.edu.pl (A. Palczewski).

https://doi.org/10.1016/j.ribaf.2022.101872
Received 19 May 2022; Received in revised form 29 September 2022; Accepted 30 December 2022
Available online 3 January 2023

and HFT interchangeably (Hendershott and Riordan, 2013). This ambiguity is due to the lack of clear definitions, as was already noted by the Securities and Exchange Commission in its concept release (SEC, 2010). The SEC acknowledged only that HFT is a subset of AT and referred to HFT as "strategies that generate a large number of trades on a daily basis". On the other hand, a rigorous definition of high-frequency traders, such as in the paper by Kirilenko et al. (2017), referring to high volume, low intraday inventory, and low end-of-day inventory restricts the possibility of analysis to rarely available data containing the identification of individual traders. The newer distinction proposed by the European Securities and Markets Authority defines high-frequency algorithmic trading as an AT technique that is characterized by the infrastructure intended to minimize latency and high intraday message rates that represent orders, quotes, or cancellations (ESMA, 2021). The CFTC Technology Advisory Committee adopted a broader understanding of high-frequency trading as a form of automated trading that uses low-latency algorithmic technology to initiate and cancel a high number of orders that could only be executed by a computer (CFTC, 2012). Taking into account: (i) the decreasing differences between HFT and other algorithmic trading participants as noted by European Securities and Markets Authority, (ii) the lack of consensus on a single definition due to the diversity of formulations as observed by Gomber et al. (2011), and (iii) the use of the two terms interchangeably in many professional and academic studies, we adopt in the rest of the paper the name high-frequency trading (HFT) for all these activities that perform automated trading without human intervention, which are characterized by order submission, cancellation, and transaction execution in a very short time.

The rapid development of high-frequency trading has stimulated an extensive debate on the advantages and disadvantages of HFT (Lewis, 2014). Together with the increase in high-frequency activity, we have observed that liquidity has also improved significantly in global equity markets. Based on these two parallel trends, one can speculate that HFT may be responsible for the increase in market liquidity. However, in the finance literature, there is no unequivocal opinion on the impact of high-frequency trading on the stock market quality. Most research indicates that HFT improves liquidity by reducing price inefficiencies and creating more trades, placing human investors in a disadvantageous position due to their reduced bargaining power (Hoffmann, 2014). However, a significant strand of research suggests a negative role of high-frequency trading on market quality. An explanation for these differences may be that computerized trading can trigger different reactions from different groups of market participants, and these reactions change over time and depend on the stock market.

Our goal in this paper is to examine, using a variety of models, the dependence of market liquidity on high-frequency activity. For some of these models, the published results indicate a positive impact of HFT on market liquidity, and for others, a negative impact is reported. To understand this ambiguity and clarify the role of HFT in stock markets, we compare these models on the same set of order-level data. We expect to find the relationship between high-frequency activity and market liquidity which remains valid independently of the model used. Furthermore, we analyze how this relationship is changing over time by investigating these changes in the period 2013–2019. Finally, we analyze the sensitivity of the results to changes in the HFT metric. To this end, we repeat the computations for the investigated models, replacing the original HFT measure of every model with the measures used in the other models. To our knowledge, no studies have compared different econometric models in such tests. Therefore, the paper addresses a long-standing literature gap.

The study is performed on publicly available XDP data from the Warsaw Stock Exchange (WSE). The data are time-stamped to the microsecond and contain in one file every message to post, modify, or cancel a market or limit order and every trade message with the trade's complete information. We collect data for the 2013–2019 period, but only for stocks that are sufficiently liquid to guarantee the daily number of messages suitable for statistical analysis (we excluded data for the years 2020–2021 for obvious reasons of the COVID-19 pandemic).

We are contributing to the existing literature in several ways. We review a broad spectrum of empirical studies and select several models suitable for investigating the influence of HFT on market liquidity. As mentioned above, some of these models reveal a positive impact of HFT on market liquidity, indicated by lower quoted spreads and higher depth in the limit order book, whereas other models present the opposite effect. One can expect that these differences result from the differences in datasets used (data from different time periods and/or different stock exchanges). But the results obtained for the analyzed models with a single set of data from the WSE give, in the majority of cases, similar results as the original research: if in the original research the model indicated a positive impact of HFT activity on market liquidity, the same effect is observed for our data, and if the effect was negative it remains negative in our computations. This finding emphasizes that the size of the capital market or the time period of observations are not the most significant factors in the relationship between market liquidity and HFT. It seems that this relation depends to a great extent on the models used in research, and a more subtle analysis is required.

Since in all models we use the same set of liquidity measures, the differences mentioned above can be caused either by the type of estimators used in regression (fixed-effect panel regression, 2SLS, or GMM estimators) or by a choice of model parameters. Among the model parameters, we distinguish between variables controlling market conditions and explanatory variables (HFT measures). To discover the source of the differences, we performed computations for each model replacing the original explanatory variable of the model with the HFT measures used in the other models. The regression coefficients for a fixed HFT measure in the different models are of the same sign. On the other hand, the results obtained for different HFT measures in a fixed model are different. These observations exclude the dependence of the results on regression estimators or variables controlling the market, indicating the dependence on explanatory variables. Thus, an important conclusion of our study is that different HFT measures estimate different aspects of high-frequency activity, and therefore these different aspects of activity affect liquidity measures differently. Our main observation is the difference between the results obtained for a measure of trade intensity (only one measure in our test) and measures of order submission. For trades, we have an indicator of the participation of algorithmic traders, a direct measure of algorithmic activity. The increase in this measure implies the improvement of market liquidity in all analyzed models. But for orders placed in the limit order book (LOB), we have no information if orders are sent by algorithmic traders. Following the approach of other researchers, we consider measures that are proxies for algorithmic/high-frequency activity. For such measures, we observe that an increase in high-frequency activity leads to a decrease in market liquidity. Hence, the conclusion is that an increase in trades executed by high-frequency traders improves market liquidity and an increase in orders sent by high-frequency traders gives the opposite effect.

The difference in market liquidity under the increase in high-frequency activity measured by the flow of transactions and the flow of orders placed in the LOB follows from separate regression analyses applied to two types of exogenous variables. To understand better how high-frequency traders influence market liquidity, we designed a test in which we observed how market liquidity was changing with the change of high-frequency activity measured simultaneously by two measures: one, corresponding to executed trades, and another, corresponding to order placement. The results of this test confirm our preliminary observations from the regression analysis. The lowest liquidity is observed for a higher number of orders generated by high-frequency traders and a lower number of transactions executed by high-frequency traders. At the same time, the highest liquidity corresponds to a low number of orders and a high number of trades by high-frequency participants. This finding is an unexpected outcome of our analysis.

The phenomenon described above is not mentioned in the existing literature, and our research provides an important insight into the relationship between HFT and market liquidity. It is clear from our analysis that an increasing number of trades positively influence liquidity. There are also papers confirming our findings that an increase in order flow metrics leads to lower market liquidity indicated by higher quoted spreads and lower depth in the limit order book (cf. Cartea et al., 2019; Van Ness et al., 2015). But there are claims that submissions that are canceled in a few seconds can improve market quality by narrowing the spreads and increasing depth, as is documented by Hasbrouck and Saar (2013) in their analysis of the NASDAQ data. This observation is confirmed by many authors who underline that high-frequency trading improves market liquidity (Hendershott et al., 2011; Lyle et al., 2015). Our results clearly show that, for the WSE data, an increase in order generation by HFTs decreases market liquidity. Our preliminary hypothesis is that the decrease in liquidity is caused by high quote volatility resulting from the frequent placement and cancellation of orders by high-frequency traders. To better understand the deep interplay between order flow and trades generated by HFT participants, we need proprietary data that enable the direct identification of not only trades, but also orders placed by HFTs.

Our research makes a new contribution to the investigation of HFT activity in emerging markets. There is a limited number of research of this type, and, to our knowledge, the present research is the first one for the Warsaw Stock Exchange. We compared regression models used in developed markets on a WSE dataset and discovered that the impact of HFT on market quality depended on the used HFT measure. But our analysis of the impact of HFT activity on market microstructure seems to be valid in developed markets as well. Our findings emphasize that different HFT measures estimate different aspects of high-frequency activity and therefore these different aspects of activity affect liquidity measures differently.

The rest of this paper is organized as follows. Section 2 reviews the literature, while Section 3 describes the liquidity and HFT measures used in our empirical analysis. Section 4 presents the methodology for estimating the effect of HFT on market quality. Section 5 discusses the results. Finally, Section 6 concludes with some discussion of the implications of the findings.

2. Literature review

Several theoretical models describe the impact of HFT on market quality. In the model of Biais et al. (2015), the authors claim that HFT helps find counterparties in transactions, which has a positive effect on trade. On the other hand, HFT generates adverse selection costs. In the (Jovanovic and Menkveld, 2016) model, high-frequency traders can update limit orders quickly and avoid some adverse selection. As a result, HFT can provide some benefits to investors by offering the seller a cost-free route to the buyer. They show that more gains from trade are realized this way. Ladley (2020) indicates that the regulations proposed to reduce the negative effects of HFT are generally ineffective in improving social welfare and market quality. The author claims that the trade-off between the complexity of an investment strategy and the time taken to its implementation is crucial. Therefore, high-frequency traders adopting the fastest strategies generate a large number of low-profit trades, while slower traders with more sophisticated strategies identify a smaller number of high-profit trades. Cartea and Penalva (2012) assume that HFT constitutes an unnecessary additional layer of intermediation between buyers and sellers and find that HFT reduces welfare. In all of the above-mentioned papers, the common observation is that HFT may increase adverse selection, which can decrease liquidity.

Empirical research on algorithmic/high-frequency trading can be roughly split into two categories. The first category uses proprietary datasets that allow direct identification of trading by HFTs. This approach has the benefit of offering deeper insights by allowing the analysis of the behavior of different types of traders and the associated effects on market outcomes. The second category uses anonymized data and empirical proxies based on indirect measures of HFTs to examine the impact of computerized trading activity on market microstructure.

A large strand of empirical papers concludes that increased high-frequency activity improves market quality. For example, Hendershott et al. (2011) examine the time series of HFT and liquidity changes for a sample of NYSE stocks and show that, especially for large stocks, HFT narrows spreads, limits adverse selection, and reduces trade-related price discovery. The study proves the positive impact of HFT on market liquidity: HFT reduces the costs of trading and increases the informativeness of the quotes. The positive impact of HFT activity on reducing the price spread of the Australian Stock Exchange is also argued by Frino et al. (2014). Similarly, Brogaard et al. (2014) and Carrion (2013), using high-frequency data on NASDAQ, find that HFTs are both demanding and supplying liquidity. In the same vein, Hagströmer and Nordén (2013) using data on 30 stocks from NASDAQ-OMX Stockholm demonstrate that the market-making HFTs increase market liquidity. Hruška (2016) analyzes the most traded stocks on the German Stock Exchange in April–October 2015 and confirms the relevance of HFTs in creating liquidity. Bellia et al. (2020) investigate HFT in the pre-opening and opening auctions, and continuous trading period on the Asia-Pacific stocks exchanges during the period April–May 2013 and show that HFT contributes to the price discovery process. However, high-frequency traders who actively contribute during the pre-opening period do not ensure price discovery during the continuous trading period.

Similar results are obtained by Abreu (2022) who, using a sample of 120 stocks listed on NASDAQ and NYSE, confirms that HFT improves market liquidity. This impact is greater for small-cap stocks. Analyzing stocks on NASDAQ, Li (2021) also finds a positive effect of HFT on market quality. Ding et al. (2021) verify the importance of the liquidity provision scheme (LPS) at NASDAQ-OMX Stockholm (NOMX) for market liquidity. The authors note that LPS provides liquidity improvements by reducing order processing costs on large-cap stocks in the NOMX and Chi-X markets. In addition, the scheme stabilizes market liquidity as market makers' decisions to supply or demand liquidity become less fragile to order imbalance. Le et al. (2021) analyze the interaction between dynamic limit order placement and market quality around two system updates introduced by the Australian Securities Exchange in 2006 and 2010. Their results indicate that, for large-cap stocks, high-frequency trading provides liquidity and stabilizes the price, when short-term volatility is high. However, they note that system upgrades can adversely affect market quality, and regulators must be prepared for market stresses.

At the same time, many studies provide evidence of the negative effects of HFTs on market liquidity. For instance, Cartea et al. (2019) use order and trade messages sent to NASDAQ, in the years 2007–2015, and find that an increase in HFT leads to lower liquidity measured by higher quoted and effective spreads and lower depth in the LOB. The above research is in line with (Van Ness et al., 2015) who study differences in cancellation activity across several US exchanges, between 2006 and 2010, and demonstrate that order cancellations by high-frequency traders are detrimental to market quality. They highlight that price spreads become wider as the cancellation rate increases. HFT provides liquidity to the thick side of the order book and demands liquidity from the thin side. These results are confirmed in the paper by Goldstein et al. (2017) examining the microstructure of the Australian Securities Exchange and highlighting that HFT provides liquidity to the thick side of the order book and demands liquidity from the thin side. Similar results are obtained in the analysis of the order book and trade data for stocks in the S&P/ASX100 index by Kwan and Philip (2015) who emphasize that non-HFT trading costs increase relative to HFT costs.

An interesting critique of the changes generated in the capital market by the introduction of algorithmic trading is carried out by Aitken et al. (2018). Their work defines fairness and market efficiency and analyzes the explosive growth of algorithmic trading between 2004 and 2011 on the London Stock Exchange and Euronext Paris. Their results show that the increase in AT improves fairness and market efficiency. The authors show that since algorithmic investors are more effective in updating their information sets and incurring lower adverse selection costs compared to other market participants, their activity leads to the improvement in market quality. However, the authors note that the positive relationship exists only for the most liquid stocks (the highest quintile). No clear conclusion on the impact of AT/HFT is also confirmed by Desagre et al. (2022). These authors find an overall improvement in liquidity on Euronext between 2002 and 2006; however, they emphasize that stocks most exposed to fast trading show the weakest liquidity growth and lose the liquidity advantage observed before the advent of algorithmic trading. The findings of Jain et al. (2021) present a new aspect of the impact of algorithmic traders on liquidity provision. The authors verify the importance of algorithmic trading in a sample of all NYSE common stocks during the period 2001–2005 and show that algorithmic traders reduce their role as liquidity providers during periods of high information asymmetry.

While for advanced economies the importance of HFT for the market microstructure has been discussed for many years, this problem is still not adequately investigated for capital markets in developing countries. Ekinci and Ersan (2018) investigate the sample of 2015 data from Borsa Istanbul and highlight that HFT activity is concentrated on a relatively smaller part of the market, mainly in the stocks with the largest market capitalization. Their research highlights that HFT has limited market relevance due to the costs of algorithmic strategies and the required advanced technology. As a result, the paper does not give a clear opinion on whether HFT improves market quality or destabilizes it. In the next study, Ekinci and Ersan (2022) focus on 30 blue chips in Borsa Istanbul for the period between December 2015 and March 2017 and find the negative impact of high-frequency activity on market quality despite its small role in the market. The authors emphasize that liquidity provision by non-HF traders is significantly reduced with the increase in high-frequency trade. Chancharat and Phuensane (2021) analyze the role of high-frequency activity on the Thailand Stock Exchange in the period January 2016 to June 2018 and confirm a negative relationship between high-frequency messages and effective spread. They indicate that the effective spread tends to be narrow when high-frequency activity is increased. The paper by Dubey et al. (2021) analyzes data for stocks from the NIFTY 50 index on the National Stock Exchange of India. A unique future of the used dataset is the identification of orders coming from algorithmic traders. Contrary to Ekinci and Ersan, the authors of this paper find undoubtedly that the rise in algorithmic trading leads to a significant improvement in the market liquidity and volatility, particularly on large-cap stocks, but the price discovery is not significantly affected.

3. Data and variables

3.1. Data

Our study focuses on stocks that are sufficiently liquid to guarantee the daily number of messages suitable for statistical analysis. We chose 20 stocks in the WIG20 index of the most actively traded stocks on the Warsaw Stock Exchange, over the period April 2013–March 2021. After examining historical price movements across all periods and taking into account market liquidity, we decided to limit our final sample to one month each year. We chose March due to the lack of significant breaks in the operation of the stock exchange caused by holidays and the settlement of futures contracts on the WIG20 index in that month. This guarantees a stable trading and no liquidity gaps on the stock exchange. Data for the years 2020–2021 were excluded, as the market liquidity in

Table 1
Orders and trades descriptive statistics only for stocks in the WIG20 index. Trading volume is expressed in millions of PLN. All sizes are divided by 1 000 000.

	N. Ord.	N. Can.	N. Shares	N. Trad.	Vol. Trad.
March 2014	10.985	5.356	19683.74	0.612	11839.58
March 2015	8.310	4.028	12409.76	0.576	9268.97
March 2016	13.195	6.422	26598.61	0.832	8792.04
March 2017	18.689	9.160	24036.63	0.882	11578.76
March 2018	16.155	7.883	22152.96	0.893	10095.77
March 2019	11.525	5.620	15898.21	0.649	7507.99
Mean	13.143	6.411	20129.99	0.740	9847.18
Daily mean	0.611	0.298	936.28	0.034	458.01
Total	78.860	38.469	120779.91	4.444	59083.11

these years was disturbed by the COVID-19 pandemic. On March 9, 2019, the WSE changed the tick size of share prices, allowing the tick size of 0.0001 PLN. After that date, the tick size depends on the stock price and trade volume and affects stocks with low prices. The effect of this change in tick size is not observed in our analysis because the change is potentially important only for one stock in our data (the results for the years 2018 and 2019 are very similar). We use publicly available XDP data. The data are time-stamped to the microsecond and contain in one file every message to post, modify, or cancel market or limit orders, and all trade messages. The descriptive statistics for our sample are presented in Table 1. In computing these statistics, only stocks present in the WIG20 index in a given period are taken into account. Therefore, for each March in the years 2014–2019, we analyze a slightly different set of stocks as only nine companies remain permanently in the index for the entire analyzed period.

The total number of orders amounts to 78.86 million over the sample period of six months (one month in each year) and 4.44 million trades with PLN 59083 million trading volume. The average number of shares entered in one month is 20130 million and ranges from a minimum of 12409.8 million in 2015 to a maximum of 26598.6 million in 2016. For each sample month, there is an average of 13.1 million orders (submitted, modified, or canceled), resulting in 740 thousand trades and 6.4 million cancellations. This large number of orders compared to the number of trades reflects the main characteristic of the high-frequency activity.

The selected sample of data is prepared for subsequent regression analysis: each trading day is divided into sub-intervals of 5-min length, and relevant quantities are computed for each sub-interval separately; all time intervals in which there are no transactions, no buy or sell orders, or no quote updates are excluded from the sample. The removed data do not influence our results, as we deleted less than 0.3% of all data. On the other hand, such prepared samples enable us to perform all computations on the same set of data. The data are winsorized, and the winsorization is calculated at the 0.5th and 99.5th percentile for data on each stock, variable, and year.

3.2. Measures of high-frequency activity

Publicly available stock exchange data usually do not distinguish orders placed by computer software from orders placed by humans. Researchers use various proxies to identify the level of high-frequency trading. In the literature, one can find many measures used successfully by various authors that correlate to a great extent with high-frequency activity. We chose several of these measures to analyze the impact of HFT on market liquidity. We selected only those measures which are computable with our dataset.

We begin with the measure introduced by Hendershott et al. (2011). These authors measured high-frequency activity by the negative value of the dollar traded volume per electronic message (measured in $100) computed in each interval Δt (in our computations we replaced dollars with the local currency). The authors called this measure *algo_trade* (ALT). In the paper by Cartea et al. (2019) the authors concentrated on the activity called *ultra-fast*, which was defined as the number of limit orders posted, and subsequently canceled within 100 ms. This measure is called PC100. Van Ness et al. (2015) defined *the cancellation rate* (CR) measured as the number of shares canceled divided by the number of shares entered in the LOB. Conrad et al. (2015) defined as the main measure of high-frequency activity *the quote updates* (QUP) – the number of changes that occur in the best bid/ask price, or in the quoted sizes at these prices, within a specified time interval Δt. Ben Ammar et al. (2020) to quantify high-frequency activity introduced *the quote-to-trade ratio* (QTR) defined as the number of quote updates at the best bid/ask price divided by the number of trades.

Ekinci and Ersan (2018) proposed a two-step approach to detect high-frequency activity: in the first step, they labeled an order as HFT if there were at least two messages corresponding to that order in one second. In the second step, they searched for orders "related" to the already detected HFT order. As the related order, they considered the order within the same second, with the same size, direction, and for the same stock. They assumed that the original HFT order and the related order were submitted by the same high-frequency trader (using the broker's ID provided by the exchange, the authors confirmed the accuracy of 97.33% of their approach). An HFT order for which a related order was found was called confirmed. Only such "confirmed orders" were used to measure high-frequency activity. Following the methodology of confirmed HFT orders, Ekinci and Ersan (2022) defined the *HFT-ratio* computed as the ratio of the volume of confirmed HFT orders divided by the volume of all submitted orders in time interval Δt. The methodology of Ekinci and Ersan (2018) was limited by the data of Borsa Istanbul which are time-stamped to the second. Since our data were time-stamped to the microsecond, we modified the above methodology in the spirit of Hasbrouck and Saar (2013) labeling orders in the first step as HFT if there were at least two messages corresponding to that order within 100 ms. In

Table 2
Descriptive statistics of high-frequency activity measures in the years 2015 and 2019. ALT denotes algo_trade: a negative value of the dollar traded volume per electronic message; PC100 is the number of limit orders posted, and subsequently canceled within 100 ms; HR denotes HFT ratio: volume of HFT labeled orders divided by the total volume of orders; HFPR is the high-frequency participation rate: the number of HFT trades divided by twice the total number of trades; CR is cancellation rate measured as the volume of canceled shares by the volume of entered shares; QTR denotes quote-to-trade ratio: the number of quote updates divided by the number of trades; QUP denotes the quote updates: the number of changes in the best bid/ask quote, or in the quoted sizes at these prices.

	Mean	SD	Min	Max	Skewness
2015	Num obser. = 28550				
ALT	−12.63	32.34	−1951	0	−26.70
PC100	20.83	128.9	0	9136	26.57
HR	0.05	0.08	0	0.96	4.28
CR	1.00	0.66	0.01	42.68	37.99
QTR	4.55	43.50	0.09	5578	87.46
QUP	52.42	168.3	1	11157	23.77
2019	Num obser. = 29446				
ALT	−6.79	13.79	−1164	0	−34.27
PC100	19.10	29.20	0	616	4.74
HR	0.03	0.04	0	0.71	2.76
CR	1.00	0.44	0	35	38.30
QTR	4.90	5.91	0.17	221	10.88
QUP	76.65	77.80	1	1062	2.93
HFPR	0.36	0.12	0	0.50	−0.92

the second step, we considered an order as a related order if it was submitted within 100 ms from the second message of the order labeled as HFT. With the above modification of confirmed orders, we computed the *HFT-ratio* (HR) of Ekinci and Ersan (2022).

All of the above measures of high-frequency activity use data in which algorithmic traders are not identified. Starting in 2018, the Warsaw Stock Exchange publishes the Algorithmic Indicator informing on transactions executed by firms engaged in algorithmic trading. Using this new indicator, we can calculate the measure introduced by Jarnecic and Snape (2014). These authors had access to nonpublic data that allowed them to examine the activity of high-frequency participants directly. They defined *the high-frequency participation rate* (HFPR) as the number of trades executed by high-frequency participants divided by twice the total number of trades.

The above-described measures are designed to indicate activity generated by computer algorithms, but the corresponding papers use different names for the activity these measures are intended to evaluate. Some papers claim to measure high-frequency activity, but others mention only indirect connection to algorithmic or high-frequency activity. Only in the paper by Hendershott et al. (2011), do the authors consider the introduced measure (algo_trade) as a direct indicator of algorithmic activity. Among the mentioned measures only the indicator of ultra-fast-activity PC100 and the HFT-ratio HR measure directly high-speed activity which can be attributed to high-frequency traders. The other indicators (cancellation rate, quote updates, and quote-to-trade ratio) measure the speed of inflow of different categories of orders. High values of these indicators prove that the average latency of traders is low, but not that they are high-frequency traders. Following the approach presented in the Introduction, we consider all selected measures as proxies of high-frequency trading. We are additionally encouraged to this slight abuse of terminology by the results of some papers that report the similarity in the impact of high-frequency activity and algorithmic activity on market liquidity (cf. Cartea et al., 2019, who obtained similar results for ALT and PC100).

The descriptive statistics for the above measures of high-frequency activity for the years 2015 and 2019 are presented in Table 2 (results for the whole data period 2014–2019 are very similar and limiting the presentation only to these two years makes the table more readable). As is visible in Table 2, the ALT measure indicates the increase in high-frequency activity changing from −12.63 in 2015 to −6.79 in 2019. This is in line with the findings by Hendershott et al. (2011). The HFT activity measured by PC100 is relatively stable. Its mean value is close to 20.00 in both years of analysis, with a downward trend in the standard deviation from 128.9 in 2015 to 29.20 in 2019. These results are significantly lower compared to the NASDAQ market analyzed by Cartea et al. (2019). The cancellation rate (CR) shows comparable mean values in both years. Van Ness et al. (2015) who developed the CR measure found very similar results. Our analysis of the WSE data shows an increase in the QTR measure from 4.55 in 2015 to 4.90 in 2019. In this case, the values are significantly different from those of Ben Ammar et al. (2020), where the average value of the QTR measure for the CAC 40 stocks listed on Euronext in 2016 is 28.59. The statistics show a strong change for the QUP measure, suggesting that the number of changes that occur in the best bid/ask price increased (52.42 vs. 76.65). This is clear evidence of increased high-frequency activity in the market. Our results are significantly lower compared to those reported by Conrad et al. (2015) on the NYSE between 2009 and 2011. The mean value of the HFT ratio (HR) calculated in the WSE is about 3%–5%. The ratio is twice higher than presented by Ekinci and Ersan (2022) on the Borsa Istanbul equity market. This difference proves that high-frequency activity is much higher on the WSE than in the Borsa Istanbul. Finally, we obtain the mean value of the high-frequency participation rate (HFPR) at the level of 0.36 and the standard deviation of 0.12, respectively. The value of this measure is reported only for the year 2019.

Table 3
Descriptive statistics of market liquidity measures in the years 2015 and 2019: quoted spread (QS), effective spread (ES), depth at the best quote (DB), depth within 10 bp of the best bid and ask quote (DX), 5-min realized spread (RS5), and 5-min adverse selection (AS5).

	Mean	SD	Min	Max	Skewness
2015	Num obser. = 28550				
QS	16.04	12.02	1.06	102.9	1.66
ES	7.48	6.2	−0.73	56.48	2
DB	139905	115737	2981	978880	2.19
DX	975455	967880	14422	4726700	1.2
RS5	3.38	10.03	−52.28	55.99	−0.11
AS5	4.14	8.68	−28.91	69	1.9
2019	Num obser. = 29446				
QS	11.34	6.12	2.15	45.45	1.42
ES	5.05	3.19	−4.82	38.36	2
DB	137245	88720	10408	817484	1.77
DX	1116724	835588	67198	4070130	0.86
RS5	0.51	8.92	−47.12	46.64	−0.71
AS5	4.55	8.85	−38.15	56.2	1.25

3.3. Measures of market liquidity

To analyze the market impact of HFT activity, we selected a universal set of liquidity measures that characterize market microstructure. For our analysis, we compute the values of these measures for each stock and each time interval Δt:

QS — quoted spread computed as the time-weighted average, over Δt, of $(a_t - b_t)/m_t$, where a_t is the best ask, b_t the best bid, and m_t the midprice at time t.

ES — effective spread computed as the value-weighted average (weighted by the transaction volume), over Δt, of $q_t(p_t - m_t)/m_t$, where q_t is a trade indicator variable that equals = 1 for a buyer-initiated trade and −1 for a seller-initiated trade, p_t is the trade price, and m_t is the quote midpoint prevailing at the time of the trade.

DB — depth at the best quote calculated as the time-weighted average of the sum of the total value of orders resting on the LOB at the best bid and ask.

DX — depth at 10 bp calculated as the time-weighted average of the sum of the total value of orders resting on the LOB within 10 basis points of the best bid and ask.

RS5 — 5-min realized spread computed as the value-weighted average of

$$q_t(p_t - m_{t+5\mathrm{min}})/m_t.$$

AS5 — 5-min adverse selection computed as the value-weighted average of

$$q_t(m_{t+5\mathrm{min}} - m_t)/m_t.$$

The descriptive statistics for the above liquidity measures for our sample are presented in Table 3.

Not all papers we are going to discuss are using each of the above measures, but a certain subset of these six measures is computed in every paper. In some papers, depth measures are defined in a slightly different way. We will indicate these differences in the model descriptions in the following section.

From Table 3, it can be seen that market liquidity increased in the analyzed period. A significant difference is seen between the mean value of the quoted and effective spreads in 2015 and 2019 — QS 16.04 vs. 11.34 and ES 7.48 vs. 5.05, respectively. Furthermore, the results show a significant decrease in the volatility of the QS and ES measures. Average DB changes from 139905 to 137245, while average DX changes from 975455 to 1116724 per five years. The depth measured by DB is relatively stable, but we observe an increase in DX that confirms the improvement of liquidity. Similarly, the average realized spread (RS5) changed from 3.38 in 2015 to 0.51 in 2019. In our study, the average value of the 5-min adverse selection (AS5) is 4.14 in 2015 and 4.55 in 2019.

4. Methodology

Analyzing the impact of high-frequency activity on market liquidity, the majority of authors consider regression models associating various measures of liquidity with measures of high-frequency activity. But the models differ substantially in the way in which the effect of endogeneity is taken into account to exclude the possible opposite causality. Another difference is in the selection of variables controlling the market conditions. The control variables used by different authors create some difficulties in our comparative analysis. Since the data used in analyzed research differ in many aspects from our data, we have to adjust the control variables to available data. Below we carefully explain which control variables are used in the original paper and what replacements we select in case we are unable to reproduce the original variables.

The simplest model of the relation between market liquidity and high-frequency activity is a panel regression model. The use of panel regression is induced by the type of data that contains information for many companies over a long period of time. The available data suggest applying the fixed-effect panel regression. A typical model of such regression has the form

$$L_{i,t} = \alpha_i + b_1 A_{i,t} + b_2 X_{i,t} + \sum_j \delta_j D_{i,j} + \varepsilon_{i,t}, \tag{1}$$

where $L_{i,t}$ is one of liquidity measures for asset i at time interval t, α_i are asset dummy variables that capture stock-specific fixed effects, $A_{i,t}$ is a measure of high-frequency activity, $X_{i,t}$ represents variables controlling market conditions, and $D_{i,j}$ is a set of time-of-trading-day dummy variables.

Panel regression raises the problem of causality of different market characteristics. Taking different measures of liquidity, we would like to identify regressors that influence changes in these measures. Unfortunately, for several reasons, regressors that are easy to access are correlated with disturbances. Such regressors are not valid explanatory variables, as there is no consistent way to estimate their effect on dependent variables. This difficulty is overcome by introducing instrumental variables, i.e., variables which are correlated with the original regressors but, on the other hand, are not correlated with error terms.

The fixed-effect panel regression is used by Hendershott et al. (2011) to analyze a sample of the NYSE data from February 2001 through December 2005 taking as the time interval Δt one day and averaging over all days of the month. The following parameters are taken as control variables: share turnover, volatility, the inverse of share price, and log market capitalization. Contrary to the original paper where the authors use as the volatility measure $\ln(H/L)$, where H, L are high/low daily prices, we take $|\ln(m_{i\Delta t}/m_{(i-1)\Delta t})|$ where $m_{i\Delta t}$ is the midquote at the end of the time interval $i\Delta t$. The authors perform a regression analysis for the following liquidity measure: quoted spread (QS), effective spread (ES), quoted depth which is our depth at the best quote (DB), 5-min realized spread (RS5), and 5-min adverse selection (AS5). They also compute regressions for 30-min realized spread and 30-min adverse selection — the computations we are not reporting. To account for endogeneity, the authors adopt as an instrumental variable "autoquote", the new exchange mechanism of automatic dissemination of a liquidity quote after every change in the limit order book (this instrument has only two states: "0" — before "autoquote" was introduced and "1" — after the introduction of "autoquote"). Since "autoquote" is a specific instrument applicable only to the NYSE data in a particular time interval, we apply as an instrumental variable the one-period lagged ALT variable.

The fixed-effect panel regression is also applied by Cartea et al. (2019). In this paper, the authors use the NASDAQ data for March in the years 2007–2015. They divide each trading day into 1-min intervals computing relevant market parameters as the interval averages of each limit order/trade event. In these computations, they use as control variables: market order imbalance calculated as a difference between the buy volume and the sell volume in the limit order book, and volatility calculated as $|\ln(m_t/m_{t-1})|$ where m_t is the midquote at the end of minute t. The authors perform a regression analysis for the following liquidity measures: quoted spread (QS), effective spread (ES), depth within 1 bp of the best bid and ask quote — we replace that measure with our depth at the best quote (DB), depth within 10 bp of the best bid and ask quote (DX). To control for endogeneity, the authors use the one-period lagged endogenous variable PC100. As the second choice, they follow the approach by Hasbrouck and Saar (2013), who argue that the effect of HFT on the market quality of one stock should be unaffected by unrelated stocks. Therefore, the authors use, as the instrument for stock i, PC100 averaged over all unrelated stocks in the sample ("unrelated" means the stock not in the same industry or index). Since our sample includes only stocks from the WIG20 index, we are unable to implement this particular instrument and limit our analysis to the instrumental variable defined by the lagged PC100 variable.

Ekinci and Ersan (2022) apply pooled panel regression to Borsa Istanbul data from December 2015 through March 2017 and compute the relevant parameters taking as the time interval Δt one day. The authors adopt several regression models proposed by Hasbrouck and Saar (2013). In Model 3 which we use for comparison, they take as control variables: trade volume, market return, and the absolute value of market return as a measure of volatility. The market return is computed as the return of the BIST 100 index. In our computations, we replace the BIST 100 index with WIG20, the corresponding index of the WSE. The authors perform regression analysis only for one liquidity measure, which they call liquidity provision (other regressions are performed for volatility and excess returns). The liquidity provision is defined as the time-weighted order volume in day t. To keep the approach of Ekinci and Ersan (2022) compatible with other results, we replace the liquidity provision with the DX market depth measure. Since Model 3 of Ekinci and Ersan (2022) is a model from the paper by Hasbrouck and Saar (2013), we extend the list of liquidity measures by measures used in this last paper, i.e., spread, which is our quoted spread (QS), and effective spread (ES).

Another approach to regression is by two-stage least squares (2SLS) estimators that use the predicted values of the regressors as explanatory instruments. In the first stage, one computes the predicted values of regressors which are used in the second stage

$$A_{i,t} = \alpha_1 + \beta_1 X^1_{i,t} + \varepsilon^1_{i,t}, \tag{2}$$

$$L_{i,t} = \alpha_2 + \gamma \hat{A}_{i,t} + \beta_2 X^2_{i,t} + \varepsilon^2_{i,t}, \tag{3}$$

where

$L_{i,t}$ — one of liquidity measures.
$A_{i,t}$ — measure of high-frequency activity estimated in the first regression equation.
$\hat{A}_{i,t}$ — values of HFT measure estimated in the first equation.
$X^1_{i,t}$ — variables controlling market conditions in the first equation.
$X^2_{i,t}$ — variables controlling market conditions in the second equation.
$\varepsilon^1_{i,t}$, $\varepsilon^2_{i,t}$ — error terms.

The intercepts α_1 and α_2 can be expanded to include dummy variables like in Eq. (1).

The 2SLS regression is used by Jarnecic and Snape (2014) to examine the liquidity supply by high-frequency traders. They analyze data from the LSE for FTSE 100 constituent stocks for the period from April 1, 2009, to June 30, 2009, and for two out-of-sample periods June 1–29, 2007, and June 2–30, 2008. All parameters are computed as averages over 5-min intervals. The control parameters in (2) are: volatility measured as $|\ln(\max(m_t)/\min(m_t))|$ where m_t is the midquote at the time interval Δt, log market capitalization, and the liquidity measure estimated in Eq. (3). For Eq. (3) the control variables are: volatility and log market capitalization the same as in Eq. (2), and inverse price. In this paper, the authors use two liquidity measures: quoted spread (QS) and quoted depth at the three best bid and ask quotes — we replace that measure with our depth within 10 bp of the best bid and ask quote (DX). In both equations, instrumental variables are included. The authors choice follows the approach of Hasbrouck and Saar (2013). In the first equation, it is the average market-wide high-frequency participation rate across all other stocks and, in the second equation, the average market-wide liquidity measure across all other stocks. Since our stocks constitute a market index, we are unable to follow that approach. Instead, we use as instrumental variables the twice-lagged values of endogenous variables (HFPR in the first equation, and the relevant liquidity measure in the second equation).

The paper by Van Ness et al. (2015) is another paper that applies a two-stage least squares model to analyze the impact of HFT on market liquidity. Contrary to the already discussed papers, this paper analyzes the data from several US market centers (27 altogether) for almost 2000 stocks in the sample period from 2006 to 2010. The authors use only monthly data provided under SEC Rule 605 (the so-called Dash-5 data). In the original paper, the regression is performed with respect to individual shares and market centers. In repeating these computations, we limit the regression to individual shares due to the type of our data. In Eq. (2), as control parameters are taken: volatility measured by the standard deviation of price, log market capitalization, log number of orders, and inverse impatient orders. For Eq. (3) the control variables are volatility, volume defined as the number of shares traded, inverse price, and average trade size. In the paper, the impatient orders are market and marketable limit orders. In our data set, the number of market orders is negligible. Taking this measure as a proxy for impatient traders will substantially reduce the sample (records with zero market orders have to be excluded). Hence, we use the number of trades as an alternative measure of impatient traders. We also replace volatility measured by the standard deviation of price with the expression $|\ln(m_{i\Delta t}/m_{(i-1)\Delta t})|$, where $m_{i\Delta t}$ is the midquote at the end of the time interval $i\Delta t$. As liquidity measures, the authors use effective spread, depth at the inside quote, size of the limit order book, realized spread, and price impact (adverse selection). Since the authors do not provide more information, we take RS5 as the realized spread, and AS5 as the price impact. The effective spread and depth at the inside quote are equal to our parameters ES and DB. We replace the size of the limit order book with DX, assuming that this is an adequate measure of the LOB depth. In Eq. (2) the authors specify two different instrumental variables. As the first instrument, they use the average cancellation rates of all stocks different than stock i, following the concept of Hasbrouck and Saar (2013). We do not use this instrumental variable for reasons already explained. The second instrument used is the twice-lagged cancellation rate, and this instrumental variable is also used in our computations.

Regression analysis can also be handled using the GMM system estimator developed by Blundell and Bond (1998). GMM estimation allows the elimination of correlations between explanatory variables and error terms by using lagged explanatory variables as instruments. Under these conditions, the resulting estimator consistently estimates the impact of an exogenous change in HFT activity on market liquidity. The GMM estimation uses the following equation

$$L_{i,t} = \alpha_i + a_1 L_{i,t-1} + b_1 A_{i,t} + b_2 X_{i,t} + \sum_j \delta_j D_{i,j} + \varepsilon_{i,t}, \tag{4}$$

where $L_{i,t-1}$ is the lagged value of a liquidity measure.

Ben Ammar et al. (2020) apply the GMM estimator to analyze the dependency of market liquidity on high-frequency activity. These authors use data for CAC 40 stocks from February 1, 2016, through May 31, 2016. From the intraday data, they compute the averages over 15-min intervals, and those averages are subjects of subsequent analysis. As control variables in Eq. (4) are used: turnover, volatility defined as the difference between the highest and the lowest price over a given time interval, and inverse price. In our calculations, we replace their volatility by the log difference $|\ln(\max(m_t)/\min(m_t))|$ where m_t is the midquote at time t. As liquidity measures, the authors use quoted spread (QS), effective spread (ES), quoted depth (DB), realized spread (RS5), and adverse selection (AS5). The difference is that their QS, ES, RS5, and AS5 are twice the values in our definitions. On the other hand, their quoted depth is one-half of our depth at best quote. The regressions for the 20-min realized spread and adverse selection performed in the original paper are not repeated in our analysis. As an instrumental variable, the authors use the relevant one-period lagged liquidity measure (cf. Eq. (4)).

A completely different approach to the analysis of the impact of high-frequency activity on market liquidity is presented by Conrad et al. (2015). The authors analyze more than 3000 stocks from different U.S. exchanges for the period 2009–2011 and the largest 300 stocks from the Tokyo Stock Exchange with two datasets: July 1, 2009, to December 31, 2009, and January 4, 2010, to March 31, 2011. The following liquidity measures are considered: effective spread, 5-min realized spread, and 5-min price impact (adverse selection). The authors also use the variance ratio and analyze the time decomposition of realized spread and price impact from one second to 20 s after the trade — but we did not repeat these computations. The impact of quote updates (QUP) on liquidity is measured as follows. All stocks are sorted into low and high update groups in each half-hour based on the median number of updates in the prior half-hour. Then the average value of a selected liquidity measure across update groups is computed by calculating a share-weighted value for each stock in a half-hour interval and then averaged across stocks in a group. Finally, the difference between the high update group and the low update group is calculated for every liquidity measure and the yearly averages of these differences are computed.

Table 4
The values in the table are coefficients of the indicated HFT measures on liquidity measures QS, ES, DB, DX, RS5, and AS5 estimated in the methodology of the paper in which the relevant HFT measure is used. Below each coefficient, we show the standard errors. Significance levels are denoted by: *** < 0.1%, ** < 1%, * < 5%.

	ALT	PC100	HR	HFPR	CR	QTR	QUP
2015							
QS	0.2136***	0.1146***	0.1007***			0.0969***	
	(0.008)	(0.006)	(0.006)			(0.006)	
ES	0.1731***	0.0835***	0.0465***		−0.3996***	0.0522***	−4.3424
	(0.008)	(0.006)	(0.006)		(0.107)	(0.007)	(6.904)
DB	−0.2635***	−0.0270***			−0.1848	−0.0035	
	(0.008)	(0.006)			(0.108)	(0.006)	
DX		−0.0257***	−0.0479***		0.2548*		
		(0.006)	(0.006)		(0.107)		
RS5	0.0766***				−0.7893***	0.0329***	−4.1164
	(0.007)				(0.100)	(0.007)	(9.752)
AS5	0.0057				0.6033***	−0.002	−0.2260
	(0.007)				(0.093)	(0.007)	(6.791)
N Ob.	28550	28550	28550		28550	26341	28656
2019							
QS	0.1757***	0.0353***	0.0364***	−0.9523***		0.1053***	
	(0.007)	(0.006)	(0.006)	(0.023)		(0.006)	
ES	0.0552***	0.0781***	0.0311***		−1.5766***	0.0307***	−2.4395
	(0.008)	(0.006)	(0.006)		(0.135)	(0.007)	(2.413)
DB	−0.3733***	−0.0567***			−1.2878***	0.0208***	
	(0.007)	(0.006)			(0.134)	(0.006)	
DX		−0.0107	−0.0360***	−0.0431*	−0.1363		
		(0.006)	(0.006)	(0.019)	(0.136)		
RS5	0.0234***				−0.7952***	0.0111	−1.6156
	(0.007)				(0.124)	(0.007)	(7.923)
AS5	−0.0097				0.3428**	−0.0051	−0.8239
	(0.007)				(0.121)	(0.007)	(7.466)
N Ob.	29446	29446	29446	29446	29446	27552	29490

5. Results

5.1. Comparison of the models

In this section, we analyze how high-frequency trading affects market liquidity measured by selected liquidity indicators. We then examine the overall impact of high-frequency activity on market quality.

We begin by repeating for our dataset the computations of the models discussed in Section 4. We perform regression calculations in the models by Ben Ammar et al. (2020), Cartea et al. (2019), Hendershott et al. (2011), Van Ness et al. (2015), and Ekinci and Ersan (2022) for the years 2014–2019. For the model by Jarnecic and Snape (2014), regression calculations are performed for the years 2018–2019 due to the lack of data for previous years. The computations for the model of Conrad et al. (2015) are also performed for the years 2014–2019. The results for all years are quite similar (some oscillations are observed, but there is no clear trend). To save space, we report in Table 4 only the coefficients of HFT measures on market liquidity measures for the years 2015 and 2019 (for the HFPR measure of Jarnecic and Snape (2014) only for the year 2019). The complete results of regression analysis with regression coefficients for control variables for all considered models are summarized in Tables 9–14 described in Appendix.

Each column of Table 4 corresponds to one of the high-frequency measures used by the authors mentioned above and described in Section 3.2. The numbers in every column are the regression coefficients of the high-frequency measure obtained by the method of the relevant paper. Below each regression coefficient, we report, in parentheses, the standard deviation for the coefficient. For the 2SLS estimators, the values in Table 4 are the regression coefficients in the second stage Eq. (3). Comparing the results of Table 4 with the results of the corresponding papers, we can observe both similarities and differences. Hendershott et al. (2011) divided stocks into groups according to capitalization. For large-cap stocks (for which the results gave the most reliable effects due to the authors), the regression coefficients for the quoted and effective spreads were negative, which was interpreted as an indicator that increased algorithmic trading activity improved market liquidity (cf. Hendershott et al., 2011, Table III). Since in our computations for ALT these coefficients are positive, their values support the opposite conclusion. On the other hand, the depth quoted in Hendershott et al. (2011) was reported with a negative coefficient. The authors claimed that the negative value of this coefficient did not contradict the positive effect of algorithmic trading on market quality as the depth decreased two times slower than the quoted spread increased.

Since in our computations, we observe that the spread increase and the depth decrease with increasing high-frequency trading, we have a clear indication that the increase in HFT harms the market liquidity. The realized spread in our computations is of the same sign as in Hendershott et al. (2011). The authors described this result as surprising and concluded that, contrary to their expectations, liquidity providers were able to capture the surplus created by the increase in algorithmic trading. But the authors claimed that this effect decreased with time. At this point, our observations are in agreement with (Hendershott et al., 2011), as we also observe that the realized spread decreases with time.

The results of Cartea et al. (2019) are in complete agreement with our results in column PC100 (cf. Cartea et al., 2019, Table III). Therefore, our calculations fully support the claim of their paper that the increase in high-frequency activity is detrimental to market liquidity. For the computations reported in columns ALT and PC100, we also checked for the endogeneity effect. Using the instrumental variables described in Section 4 we analyzed the problem of causality, but similarly, as in the original papers, we found that reverse causality is not a concern for the obtained results.

The results of Ekinci and Ersan (2022) show that an increase in HFT-ratio has a significantly negative impact on liquidity provision (cf. Ekinci and Ersan, 2022, Table 4). That result is in agreement with our computations reported in column HR of Table 4. The regression coefficient for market depth DX is negative, which is in agreement with the sign of the coefficient for the liquidity provision in Ekinci and Ersan (2022), and the coefficients for the quoted and effective spreads are positive. Thus, the signs of all these coefficients support the observation of the negative impact of HFT on market liquidity.

The calculations of the model by Jarnecic and Snape (2014) are performed only for the years 2018 and 2019, and the results for the year 2019 are reported in Table 4 in column HFPR. The regression coefficient for the quoted spread is negative and agrees with their result (cf. Jarnecic and Snape, 2014, Table 8). But the results for the quoted depth are of the opposite sign. The results of Jarnecic and Snape unanimously support the conclusion that the increase in high-frequency activity improves market liquidity. Our results are not so conclusive. We can claim that our results rather support than oppose the conclusion of Jarnecic and Snape by comparing not only the signs of the coefficients but also their size. The coefficient for the quoted spread is much larger than in their paper and the coefficient for the quoted depth is much smaller (comparing absolute values). Therefore, we can say that a positive effect on market liquidity caused by the quoted spread is more profound than a negative effect of the quoted depth.

The results of Van Ness et al. (2015) support the conclusion that the increase in the cancellation rate decreases the market liquidity. Our results reported in column CR indicate rather the opposite effect. The regression coefficients for the effective spread are negative contrary to their results (cf. Van Ness et al., 2015, Table 6). On the other hand, the coefficients for the quoted depth are also negative (this time in agreement with Van Ness et al., 2015), which indicates a positive impact of cancellations on market liquidity. But this effect is not completely clear: the coefficient for DB is negative in both 2015 and 2019, but the coefficient for DX is positive for 2015 and negative for 2019. Unfortunately, the significance of these coefficients is not very high. We cannot draw any conclusions about the realized spread and the price impact, as (Van Ness et al., 2015) regressed the absolute values of these measures. However, comparing the results in column CR with the coefficients in column ALT, we observe that the improvement of liquidity indicated by the decrease of the effective spread reduces gains of liquidity providers and increases price impacts. These effects agree with the indicated improvement in liquidity.

The findings of Ben Ammar et al. (2020) indicate that high-frequency activities are more concentrated when spreads are narrow. Our results indicate the opposite effect for QS and ES. The estimation with our data emphasizes that QTR positively affects QS with a coefficient of 0.10 and ES with a coefficient of 0.05 in the year 2015, suggesting that an increase in QTR leads to an increase in quoted and effective spreads. The value of DB increases significantly with QTR in 2019, suggesting that the increase in high-frequency activity improves the depth of the LOB. This finding is consistent with the results of Ben Ammar et al. (2020). Furthermore, QTR positively affects RS5 with a significant coefficient of 0.03, implying that in this way high-frequency traders have a better chance of benefiting from the provision of liquidity. On the other hand, we found that higher high-frequency activity corresponded to lower AS5. The obtained results are the same for 2015 and 2019. These results are similar to the results reported in Ben Ammar et al. (2020, Table 5). Therefore, Ben Ammar et al. (2020) indicate that high-frequency activity improves intraday stock liquidity, but our findings cannot confirm that higher intensity of high-frequency activity is associated with lower spreads and higher depth.

The results of Conrad et al. (2015) are difficult to compare quantitatively with our results, as these authors classified stocks in quintiles corresponding to company size and trade activity, and there are no results for the whole sample. Nevertheless, the coefficients in Conrad et al. (2015, Table 4) and in the QUP column of Table 4 are of the same sign. This indicates that the increase in the quote updates decreases the effective spread, which corresponds to the improvement in liquidity. The decline in the realized spread is in agreement with the results in column QUP and could mean the reduction of gains to liquidity providers. There is no clear explanation for the decline in the price impact, the authors are only speculating on possible reasons for this effect.

The conclusions we can draw from the results of Table 4 are ambiguous. Comparing only the regression coefficients for the quoted and effective spreads, we have positive values for the models of Hendershott et al. (2011), Cartea et al. (2019), Ben Ammar et al. (2020), and Ekinci and Ersan (2022) in the years 2015 and 2019. In all cases, the values obtained are of high significance and with small standard deviations. Thus, the increase in high-frequency activity measured by ALT, PC100, QTR, and HR increases the quoted and effective spreads. Therefore, we can conclude that the increase in high-frequency activity leads to lower market liquidity. On the other hand, the effective spread for the models of Van Ness et al. (2015) and Conrad et al. (2015) in the years 2015 and 2019, and the quoted spread for the model of Jarnecic and Snape (2014) in the year 2019 have negative regression coefficients, which lead to the opposite conclusion.

For the depth of the market measured by two measures DB and DX, the coefficients for the models of Hendershott et al. (2011) and Cartea et al. (2019) are negative, with high significance, and a small standard deviation. A similar observation applies to the depth DX for the model of Ekinci and Ersan (2022). This supports the previous conclusion that increasing high-frequency activity reduces market liquidity. For the model of Ben Ammar et al. (2020), the conclusion is not that clear: for the year 2015 we have a negative coefficient (but with low significance) and in the year 2019 a positive coefficient (with high significance). For the model

Table 5
The impact of high-frequency trade on market quality. Summary of the compared studies. The HFT measures of the first column are defined in Section 3.2 and the market liquidity measures mentioned in column 4 are described in Section 3.3.

HFT measure	Compared studies	HFT impact	Detailed description of similarities and differences in results
ALT	Hendershott et al. (2011) NYSE data	positive	decrease in spreads QS and ES, decrease in depth DB (positive impact — depth decreases slower than spread increases)
	present study WSE data	negative	increase in spreads QS and ES, decrease in depth DB
PC100	Cartea et al. (2019) NASDAQ data	negative	increase in spreads QS and ES, decrease in depths DB and DX
	present study WSE data	negative	increase in spreads QS and ES, decrease in depths DB and DX
HR	Ekinci and Ersan (2022) Borsa Istanbul data	negative	decrease in *liquidity provision*
	present study WSE data	negative	decrease in depth DX, additionally increase in spreads QS and ES
HFPR	Jarnecic and Snape (2014) LSE data	positive	decrease in spread QS and decrease in *quoted depth*
	present study WSE data	weak positive	decrease in spread QS and decrease in depth DB (weak positive impact, depth decreases slower than spread increases)
CR	Van Ness et al. (2015) NYSE, NASDAQ data	negative	increase in spread ES and decrease in depth DB and *LOB size*
	present study WSE data	weak positive	decrease in spread ES, depths DB, and DX increase or decrease depending on the year (weak positive impact)
QTR	Ben Ammar et al. (2020) Euronext Paris data	positive	decrease in spreads QS and ES, and increase in depth DB
	present study WSE data	weak negative	increase in spreads QS and ES, depth DB increase or decrease depending on the year (weak negative impact)
QUP	Conrad et al. (2015) NYSE data	positive	decrease in spread ES
	present study WSE data	positive	decrease in spread ES

of Van Ness et al. (2015) we have for the year 2015 a negative coefficient for DB and a positive for DX (both with low significance) and for the year 2019 the coefficients for both depth measures are negative. Hence, the depth of the market is rather decreasing with increasing high-frequency activity. Similar results are for the model of Jarnecic and Snape (2014) for the year 2019: the market depth measured by DX decreases with increasing high-frequency activity. It is difficult to draw a clear conclusion from the results for the realized spread and price impact, as the regression coefficients for these measures are of different signs for each model.

We can summarize the results in Table 4 as follows: for the spread measures we have four models with positive and three models with negative coefficients; for the depth measures we have four models with negative coefficients and two models in which the coefficients are of the opposite signs in the years 2015 and 2019. Only for three models (Hendershott et al., 2011; Cartea et al., 2019; Ekinci and Ersan, 2022) do the results for the spread measures agree with the results for the depth measures and indicate that the increase in high-frequency activity decreases the market liquidity. For two models (Jarnecic and Snape, 2014; Van Ness et al., 2015), the results for the spread measures show an improvement in liquidity and the results for the depth measures indicate the opposite effect. With the results by Ben Ammar et al. (2020), the situation is even more complicated: the spread measures show that an increase in high-frequency activity leads to lower liquidity and the results for the depth measures are inconclusive (the regression coefficients change signs between 2015 and 2019).

For an efficient comparison of the studies analyzed in the paper, we summarize the results in Table 5. We compare seven models using different types of high-frequency measures applied to different stock markets and replicate them for the Warsaw Stock Exchange data. The replicated results agree with the results of the original work in four cases — in one case (Jarnecic and Snape, 2014), the similarity is partial. In one case, the results for the WSE data are different from the results of the original work, and for two works there are partial differences.

5.2. Testing high-frequency measures: impact on market liquidity

The differences between our results and the results reported in the discussed papers are not surprising. The results of the selected papers were obtained for data from different time periods and for stock exchanges that differ substantially from the Warsaw Stock Exchange. Much more difficult is to explain why, for the same data, we obtain different results for different high-frequency measures. After all, these measures are designed to reflect the impact of the high-frequency activity on stock market liquidity. To explain the observed differences, we have to consider two possibilities: the results depend strongly on the applied regression estimator, or the differences are caused by the selection of variables used in regression equations. Considering the variables in the model, we should distinguish between variables controlling for market conditions and variables measuring high-frequency activity.

To investigate the source of the above-described discrepancy in results, we designed an additional test. For each model in Table 4, we run a sequence of computations that successively replace the original high-frequency measure with the other measures described in Section 3.2. The results of these computations are as follows. Excluding the model of Conrad et al. (2015) (we will discuss the

Table 6
The values in the table are coefficients of the indicated HFT measures on liquidity measures QS, ES, DB, DX, RS5, and AS5 estimated in the model of Cartea et al. (2019). Below each coefficient, we show the standard errors. Significance levels are denoted by: *** < 0.1%, ** < 1%, * < 5%.

	ALT	PC100	HR	HFPR	CR	QTR	QUP
2015							
QS	0.1557***	0.1146***	0.0606***		0.0011	0.1754***	0.0878***
	(0.006)	(0.006)	(0.006)		(0.006)	(0.006)	(0.006)
ES	0.1279***	0.0835***	0.0186**		0.0043	0.0993***	0.0607***
	(0.006)	(0.006)	(0.006)		(0.006)	(0.006)	(0.006)
DB	−0.2326***	−0.0270***	−0.0283***		−0.0413***	−0.0242***	−0.0044
	(0.006)	(0.006)	(0.006)		(0.006)	(0.006)	(0.006)
DX	−0.1226***	−0.0257***	−0.0250***		−0.0169**	−0.0220***	−0.0006
	(0.006)	(0.006)	(0.006)		(0.006)	(0.006)	(0.006)
RS5	0.0165**	0.0391***	−0.0169**		0.0012	0.0411***	0.0474***
	(0.005)	(0.006)	(0.006)		(0.005)	(0.005)	(0.006)
AS5	0.0488***	0.0033	0.0292***		−0.0017	0.0142**	−0.0159**
	(0.005)	(0.005)	(0.005)		(0.005)	(0.005)	(0.005)
2019							
QS	0.1139***	0.0353***	0.0180**	−0.0500***	0.0274***	0.1141***	−0.0174**
	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)
ES	0.0071	0.0781***	0.0317***	−0.0954***	0.0168**	0.0088	0.0334***
	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)	(0.007)
DB	−0.2655***	−0.0567***	0.0076	−0.0342***	−0.0328***	−0.0452***	−0.0548***
	(0.005)	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)
DX	−0.1611***	−0.0107	−0.0074	−0.0339***	−0.0198***	−0.0359***	−0.0003
	(0.005)	(0.006)	(0.005)	(0.005)	(0.005)	(0.005)	(0.006)
RS5	−0.0154**	0.0524***	−0.0037	−0.0342***	−0.0086	−0.0067	0.0568***
	(0.005)	(0.006)	(0.005)	(0.005)	(0.005)	(0.005)	(0.006)
AS5	0.0145**	−0.0308***	0.0124*	0.0087	0.0138**	0.0086	−0.0482***
	(0.005)	(0.006)	(0.005)	(0.005)	(0.005)	(0.005)	(0.006)

results for this model separately), the results for all analyzed regression models are very similar, eliminating the possibility that the observed differences result from the choice of the regression estimators or variables controlling for market conditions. On the other hand, in each regression model, we observe differences between the HFT measures very similar to the differences in Table 4. This observation suggests that the differences are caused by the measures of high-frequency activity used as explanatory variables. The most striking is the difference between the results for the HFPR and the remaining measures. To save space, we will discuss in detail the results for the 2SLS model of Jarnecic and Snape (2014), which is the model designed for the HFPR measure, and the fixed-effect panel regression model of Cartea et al. (2019).

Table 6 presents the regression coefficients for different high-frequency measures in the model of Cartea et al. (2019). For ALT, PC100, CR, QTR, QUP, and HR, we have positive regression coefficients for spread measures (QS and ES) and negative coefficients for depth measures (DB and DX) both in the years 2015 and 2019. The decrease in spreads and the increase in depth with increasing high-frequency activity is a clear indicator of a decrease in market liquidity. On the other hand, for the HFPR measure in 2019, we observe negative regression coefficients for the spread measures, but also negative coefficients for the depth measures. This gives an inconclusive result on the influence of high-frequency activity on market liquidity. The results for the realized spread and price impact are not conclusive. We rather observe a positive sign of RS5 and AS5 for 2015. For 2019, the sign of RS5 is in many cases negative, but the sign of AS5 is mostly positive. The results for the model by Jarnecic and Snape (2014) presented in Table 7 are very similar to the results of Table 6. The only visible difference is the sign for RS5 in 2015, which is in most cases negative.

Computations similar to those reported in Tables 6 and 7 were performed also for other analyzed models. For models of Hendershott et al. (2011), Van Ness et al. (2015), Ben Ammar et al. (2020), and Ekinci and Ersan (2022) the results are similar to those from Tables 6 and 7 — a decrease of spreads and depth for the HFPR measure and an increase of spreads and a decrease of depth for the other high-frequency measures. The only exception is the results for the CR measure in the model of Van Ness et al. (2015). In this model, we obtained similar results for the CR measure as for the HFPR measure, i.e., a decrease in spreads and depth.

The results for the model by Conrad et al. (2015) differ substantially from the results in Tables 6 and 7. We observe that increases in PC100, CR, QTR, QUP, and HR result in a decrease in the quoted and effective spreads. On the other hand, the increase in HFPR increases the quoted and effective spreads. Thus, these results indicate the impact on market liquidity, which is in the opposite direction to that following from the results of Tables 6 and 7. To explain these differences, we have to discuss in detail the methodology used by Conrad et al. (2015) in comparison to panel regression. Let us recall that the authors of that paper first computed average liquidity measures over stocks in low and high update groups in each time interval, then calculated the difference of these averages between high and low update groups, and finally computed the time averages of these differences. In the within-group panel regression, one performs averaging in the opposite order: first, the time averages of all liquidity measures and high-frequency measures are computed, then the within-group coefficient estimator is obtained by generalized least-squares.

Table 7
The values in the table are coefficients of the indicated HFT measures on liquidity measures QS, ES, DB, DX, RS5, and AS5 estimated in the model of Jarnecic and Snape (2014). Below each coefficient, we show the standard errors. Significance levels are denoted by: *** < 0.1%, ** < 1%, * < 5%.

	ALT	PC100	HR	HFPR	CR	QTR	QUP
2015							
QS	3.2320***	1.8691***	1.3988***		15.4327***	2.8197***	1.1257***
	(0.021)	(0.020)	(0.025)		(0.213)	(0.019)	(0.017)
ES	3.2080***	1.5697***	0.7269***		15.6867***	2.4286***	0.8681***
	(0.025)	(0.023)	(0.028)		(0.231)	(0.028)	(0.019)
DB	−3.2355***	−0.6126***	−0.5864***		−16.3599***	−0.6769***	−0.2393***
	(0.013)	(0.024)	(0.027)		(0.064)	(0.033)	(0.019)
DX	−1.9503***	−0.3136***	−0.2520***		−12.5864***	−0.4403***	−0.1078***
	(0.022)	(0.017)	(0.019)		(0.113)	(0.023)	(0.013)
RS5	−0.0732	−1.1989***	−1.5569***		−19.6932***	1.1848***	−0.9925***
	(0.042)	(0.025)	(0.027)		(0.209)	(0.036)	(0.019)
AS5	2.3998***	1.9009***	1.9072***		21.2877***	0.6137***	1.4049***
	(0.035)	(0.021)	(0.024)		(0.180)	(0.037)	(0.017)
2019							
QS	1.5109***	0.9450***	1.3470***	−0.9523***	20.2009***	1.8464***	0.3253***
	(0.022)	(0.019)	(0.033)	(0.023)	(0.098)	(0.024)	(0.016)
ES	0.2208***	1.1978***	1.7592***	−1.4897***	22.0194***	0.0539	0.5216***
	(0.026)	(0.019)	(0.033)	(0.022)	(0.165)	(0.031)	(0.016)
DB	−2.4585***	−0.7919***	−0.3681***	−0.0792***	−16.5140***	−0.9099***	−0.3676***
	(0.011)	(0.019)	(0.034)	(0.023)	(0.060)	(0.028)	(0.016)
DX	−1.6105***	−0.2109***	−0.4918***	−0.0431*	−15.7026***	−0.4573***	−0.0058
	(0.016)	(0.016)	(0.027)	(0.019)	(0.109)	(0.023)	(0.013)
RS5	−0.4943***	−1.4540***	−1.9196***	−0.3819***	−19.6649***	0.6410***	−1.0408***
	(0.028)	(0.018)	(0.035)	(0.026)	(0.065)	(0.033)	(0.016)
AS5	0.4967***	1.6763***	2.3188***	−0.1251***	18.2016***	−0.6427***	1.1588***
	(0.028)	(0.017)	(0.033)	(0.027)	(0.054)	(0.033)	(0.015)

This change in the order of averaging can be responsible for the differences in results. To confirm this hypothesis, we performed the computation of Conrad et al. (2015) with a change in averaging. First, for each stock and each trading day, we divided all time intervals into two groups of high and low values of the quote update measure (QUP). We then computed the time averages of liquidity measures for each stock separately over the high and low update groups and the difference between these averages. These differences were averaged (share-weighted) over all stocks. The results of these computations gave coefficients of the opposite sign to those in the QUP column in Table 4. Repeating these computations for the remaining high-frequency measures, we obtained results similar to the results in Tables 6 and 7.

Taking into account the results in Tables 6 and 7, and the results of the above modification of the methodology of Conrad et al. (2015), we can claim that the increase in ALT, PC100, CR, QTR, QUP, and HR implies the increase in liquidity measures. This result suggests that an increase in high-frequency trading leads to lower liquidity. A weak point of this conclusion is that the HFT measures, for which the conclusion is valid, are all indirect measures (proxies) of high-frequency activity. On the other hand, for the HFPR measure, which directly measures algorithmic activity, we obtain the opposite effect: an increase in HFPR leads to lower quoted and effective spreads, which suggests the improvement of liquidity. However, the results for HFPR are ambiguous: the LOB depth also decreases with increasing HFPR. Thus, we have an inconsistent result: some measures indicate that the increase of algorithmic/high-frequency trading decreases the market liquidity, but the results for HFPR suggest that the increase of algorithmic/high-frequency trading improves the liquidity.

To understand this inconsistency, let us observe that HFPR measures the algorithmic activity in trading whereas the other measures are proxies to high-frequency activity in order placing. It might be that high activity in order placing decreases liquidity, and frequent trades improve liquidity, in particular, leading to the decrease of the quoted and effective spreads. To better understand the impact of different high-frequency measures on liquidity, we performed computations in which we compared HFPR with one of the measures related to order placing (we chose PC100, but the results were similar for the other measures). The calculations were as follows. For each stock and each trading day, we divided all time intervals into two groups of low and high PC100 values using the median value as a breaking point, each of these groups was further divided into subgroups (also using the median) of low and high HFPR values. As a result of these operations, all time intervals were divided into four groups corresponding to high/low values of PC100 and high/low values of HFPR. In each of these groups, we computed the time averages of the liquidity measures. Finally, for each group, we calculated the share-weighted average liquidity measure. The results of these calculations for the year 2019 are reported in Table 8.

Inspecting the values in Table 8, we observe the highest values of QS and ES in the HL group 10.33 for QS and 4.68 for ES, respectively. In other words, a higher number of orders placed by high-frequency participants and a lower number of high-frequency trades hurt market quality. In turn, the lowest spread, which is a signal of high market liquidity, is observed in the LH group, i.e., for

Table 8
Average values of respective liquidity measures. Averages are taken over time intervals corresponding to high values of PC100 and high values of HFPR (HH), high values of PC100 and low values of HFPR (HL), low values of PC100 and high values of HFPR (LH), and low values of PC100 and low values of HFPR (LL). The values for DB and DX are divided by 10 000.

Measure	HH	HL	LH	LL
QS	10.05	10.33	9.49	9.47
ES	4.33	4.68	4.20	4.25
DB	15.11	14.96	15.92	16.17
DX	122.14	121.74	122.68	122.91

Table 9
Regression results for the model of Hendershott et al. (2011). The table presents the coefficient of algo_trade (ALT) on different liquidity measures: quoted spread (QS), effective spread (ES), depth at the best quote (DB), 5-min realized spread (RS5), and 5-min adverse selection (AS5). The control variables in this model are: share turnover (TOver), volatility (Volat), 1/price (InvPrice), and log market cap (LogMCap). Fixed effects and time dummies are included. The adjusted R^2 value is shown below the coefficients of the regression. Significance levels are denoted by: *** < 0.1%, ** < 1%, * < 5%.

reg.variable	QS	ES	DB	RS5	AS5
ALT	0.1757***	0.0552***	−0.3733***	0.0234***	−0.0097
TOver	0.1298***	0.1472***	−0.0918***	0.0845***	−0.0347***
Volat	0.1400***	0.1163***	−0.1085***	−0.4679***	0.4994***
InvPrice	0.2291***	−0.0261	0.1019***	−0.0814**	0.0644*
LogMCap	0.2804***	0.0809**	0.2025***	−0.0447	0.0674*
adj. R^2	0.1174	0.0621	0.1753	0.2036	0.2451

small PC100 and high HFPR — a low number of orders placed by high-frequency traders and a large number of trades by high-frequency traders. In this group, the spreads are 9.49 for QS and 4.20 for ES, respectively. We obtained a similar relationship in the case of liquidity measured by DB and DX, the market depth measures. Accordingly, the lowest values are in the HL column 14.96 for DB and 121.74 for DX, while the highest values are in the LL column 16.17 for DB and 122.91 for DX, but the values in the LH column are not much smaller. These results prove that market liquidity is positively affected by HFT strategies consisting of a low number of orders and a high number of trades. The opposite strategy with a significant number of orders and a low number of trades results in a decrease in liquidity.

Interpreting the differences between the results for the trade flow metric HFPR and the order flow metric PC100, we should take into account the nature of these metrics. HFPR measures all trades in which high-frequency traders participate. Compared to HFPR, the order flow metric PC100 does not measure all HFT orders but, in fact, only aggressive orders. High values of PC100 indicate that high-frequency traders place and cancel orders very frequently, increasing quote volatility. This phenomenon was first investigated by Baruch and Glosten (2013), who called it "flickering quotes". Quote volatility was further researched by Hasbrouck (2018). His study argues that induced quote volatility degrades market quality and is often attributed to the manipulative practices of quote stuffing and spoofing. Quote stuffing is a strategy in which orders are canceled and resubmitted to slow-down other traders from trading. This strategy adversely affects market quality, as empirically documented by Egginton et al. (2016). In the case of spoofing, the trader briefly exposes quotes that are not intended for execution to try to provoke quote changes by other traders (and make a profit from these changes). The practice of spoofing can also hurt market liquidity. High values of the other analyzed metrics of order flow can also be associated with the above-mentioned strategies deteriorating market quality, as is documented by Van Ness et al. (2015) for the cancellation rate (CR). Since all analyzed order flow measures can also be classified as measuring aggressive behavior of high-frequency participants, that might be an explanation for why the increase of these measures decreases market liquidity. For HFPR, Jarnecic and Snape (2014) suggest that HFTs engage in various strategies that benefit market liquidity. Thus, they claim that high-frequency participants can contribute to liquidity by increasing the volume of trade.

6. Conclusions

Today, algorithmic/high-frequency trading is a dominant practice in all developed security markets. Our analysis of publicly available data from the Warsaw Stock Exchange shows that emerging markets follow this trend. The large participation of high-frequency traders was already visible in 2013. Several measures of high-frequency activity indicate that this participation increases over time. However, similarly to developed markets where some published results show a positive effect of HFT on market liquidity, while in other publications, a negative effect is reported, our findings for the WSE data are also inconclusive.

Our paper shows in a new perspective the interplay between high-frequency activity and market quality. Different models, in particular different HFT measures used as explanatory variables, give contradictory results. The increase in ALT, PC100, CR, QTR, QUP, or HR, which indicates an increase in high-frequency activity, implies an increase in QS and ES. This result strongly suggests that an expansion in high-frequency trading leads to a reduction in liquidity. In contrast, for the HFPR measure, which directly measures high-frequency trading, we observe the opposite effect: an increase in HFPR leads to lower quoted and effective spreads, suggesting improved liquidity. However, these results for HFPR should be interpreted with caution, because the LOB depth

Table 10
Regression results for the model of Cartea et al. (2019). The table presents the coefficient of the number of limit orders posted and subsequently canceled within 100 ms (PC100) on different liquidity measures: quoted spread (QS), effective spread (ES), depth at the best quote (DB), and depth within 10 bp of the best bid and ask quote (DX). The control variables in this model are: market order imbalance calculated as a difference between the buy volume and the sell volume in the limit order book (MImb), and volatility (Volat). Fixed effects and time dummies are included. The adjusted R^2 value is shown below the coefficients of the regression. Significance levels are denoted by: *** < 0.1%, ** < 1%, * < 5%.

reg.variable	QS	ES	DB	DX
PC100	0.0333***	0.0754***	−0.0587***	−0.0116
MImb	−0.0117*	0.0057	0.0964***	0.1310***
Volat	0.1473***	0.1084***	−0.1106***	−0.1056***
adj. R^2	0.0967	0.0591	0.0962	0.1298

Table 11
Regression results for the model of Ekinci and Ersan (2022). The table presents the coefficient of the HFT ratio (HR) on different liquidity measures: quoted spread (QS), effective spread (ES), and depth within 10 bp of the best bid and ask quote (DX). The control variables in this model are: stock trading volume (TrVol), market return measured as the return of the WIG20 index (MRet), and market volatility (MVol). Fixed effects and time dummies are included. The adjusted R^2 value is shown below the coefficients of the regression. Significance levels are denoted by: *** < 0.1%, ** < 1%, * < 5%.

reg.variable	QS	ES	DX
Intercept	−0.1564***	−0.1243***	0.5583***
HR	0.0364***	0.0311***	−0.0360***
TrVol	0.0054	0.0969***	0.0791***
MRet	0.0003	0.0019	0.0017
MVol	−0.0148**	−0.0154**	−0.0043
adj. R^2	0.0721	0.0469	0.1066

also decreases with the increase of HFPR. To explain the above inconsistency, we observe that the HFPR measures high-frequency activity in trading, while the other measures are indicators of high-frequency activity in order placing. This study appears to be the first to indicate a positive impact on market liquidity by high-frequency strategies generating a small number of orders and a large number of trades. On the other hand, the market is adversely affected by activity in which a high volume of orders is placed and a low number of transactions is made. In general, these results indicate that liquidity measures depend differently on the flow of limit orders and trades. Our preliminary explanation of the differences between the results for the trade flow metric and the order flow metrics requires an extended investigation. That analysis is possible with data that provides for each message (order placing, order canceling, or trade) the information if the message is sent by a high-frequency trader. This analysis will be the subject of our future research. Our findings highlight the difficulty in investigating the relationship between algorithmic/high-frequency trading and market microstructure, indicating that the results can depend not only on stock markets or data periods, but also on the explanatory variables used in the analysis. Our results can also be helpful to market regulators suggesting a direction for modifications to the existing system of trade supervision, regulation, and organization.

CRediT authorship contribution statement

Renata Karkowska: Conceptualization, Investigation, Methodology, Resources, Validation, Writing – original draft, Writing – review & editing. **Andrzej Palczewski:** Conceptualization, Data curation, Investigation, Methodology, Software, Validation, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

Acknowledgments

The authors are greatly indebted to Piotr Kowalczyk for generous help in the baseline data analysis software. We also would like to express our gratitude to Andrzej Sopoćko from the University of Warsaw and Maciej Trybuchowski from the Central Securities Depository of Poland S.A., who helped us to obtain data from the Warsaw Stock Exchange.

Table 12
Regression results for the model of Jarnecic and Snape (2014). The table presents the coefficient of the high-frequency participation rate (HFPR) on liquidity measures: quoted spread (QS) and depth within 10 bp of the best bid and ask quote (DX). The control variables in this model are: volatility measured as a log of the high divided by low mid-quote price (VolMax), log market cap (LogMCap), 1/price (InvPrice), and twice-lagged market liquidity measure (MQ_{t-2}). The adjusted R^2 value is shown below the coefficients of the regression. Significance levels are denoted by: *** < 0.1%, ** < 1%, * < 5%.

reg.variable	QS	DX
Intercept	−0.0276	0.0276
HFPR	−0.9523***	−0.0431*
VolMax	−0.0188***	0.0207***
LogMCap	0.1251***	0.0658***
InvPrice	0.0833***	0.0602***
MQ_{t-2}	0.3822***	0.7307***
adj. R^2	0.2418	0.5325

Table 13
Regression results for the model of Van Ness et al. (2015). The table presents the coefficient of the cancellation rate (CR) on different liquidity measures: effective spread (ES), depth at the best quote (DB), depth within 10 bp of the best bid and ask quote (DX), 5-min realized spread (RS5), and 5-min adverse selection (AS5). The control variables in this model are: stock trading volume (TrVol), volatility (Volat), 1/price (InvPrice), and trade size (TrSize). The adjusted R^2 value is shown below the coefficients of the regression. Significance levels are denoted by: *** < 0.1%, ** < 1%, * < 5%.

reg.variable	ES	DB	DX	RS5	AS5
Intercept	−0.0389	0.0303	0.0266	−0.0108	−0.0037
CR	−1.5766***	−1.2878***	−0.1363	−0.7952***	0.3428**
TrVol	0.1468***	0.1394***	0.1450***	0.1172***	−0.0747***
Volat	0.2166***	−0.1355***	−0.1742***	−0.4212***	0.4815***
InvPrice	−0.0824***	−0.1851***	−0.1225***	−0.0073	−0.0212**
TrSize	−0.0468***	0.2363***	0.1335***	−0.0773***	0.0690***
adj. R^2	0.0379	0.0933	0.0565	0.2000	0.2386

Table 14
Regression results for the model of Ben Ammar et al. (2020). The table presents the coefficient of the quote-to-trade ratio (QTR) on different liquidity measures: quoted spread (QS), effective spread (ES), depth at the best quote (DB), 5-min realized spread (RS5), and 5-min adverse selection (AS5). The control variables in this model are: lagged market liquidity measure (MQ_{t-1}), volatility (Volat), share turnover (TOver), and 1/price (InvPrice). Fixed effects and time dummies are included. The value of the Wald χ^2 test is shown below the coefficients of the regression. Significance levels are denoted by: *** < 0.1%, ** < 1%, * < 5%.

reg.variable	QS	ES	DB	RS5	AS5
Intercept	−0.5124	0.3174	0.1964	0.4062	−0.3659
QTR	0.1053***	0.0307***	0.0208***	0.0111	−0.0051
MQ_{t-1}	0.3846***	0.1712***	0.4654***	−0.0199***	−0.0136**
Volat	0.0681***	0.0286***	−0.0658***	−0.4761***	0.4852***
TOver	0.0274***	0.1216***	0.0157*	0.0509***	−0.0196*
InvPrice	0.0102	−0.0483***	0.0071	−0.0015	−0.0113
Wald χ^2 test	4613***	1161***	6145***	4655***	5072***

Appendix

In Tables 9–14, we present the results of the regression analysis of liquidity measures on high-frequency trading proxies and variables controlling market quality. The analysis is carried on for the stocks in the WIG20 index of the WSE for the period from March 1, 2019, to March 31, 2019. There is a separate table for each analyzed model (each HFT measure).

References

Abreu, D., 2022. High Frequency Traders and Liquidity. Working Paper, CUNY.
Aitken, M.J., Aspris, A., Foley, S., Harris, F.H.D.B., 2018. Market fairness: The poor country cousin of market efficiency. J. Bus. Ethics 147, 5–23. http://dx.doi.org/10.1007/s10551-015-2964-y.
Anagnostidis, P., Fontaine, P., 2020. Liquidity commonality and high frequency trading: Evidence from the french stock market. Int. Rev. Financ. Anal. 69, 101428. http://dx.doi.org/10.1016/j.irfa.2019.101428.
Baruch, S., Glosten, L.R., 2013. Fleeting orders. SSRN Electron. J. http://dx.doi.org/10.2139/ssrn.2278457.
Bellia, M., Pelizzon, L., Subrahmanyam, M.G., Uno, J., Yuferova, D., 2020. Low-latency trading and price discovery: Evidence from the Tokyo Stock Exchange in the pre-opening and opening periods. SSRN Electron. J. http://dx.doi.org/10.2139/ssrn.3619823.
Ben Ammar, I., Hellara, S., Ghadhab, I., 2020. High-frequency trading and stock liquidity: An intraday analysis. Res. Int. Bus. Finance 53, 101235. http://dx.doi.org/10.1016/j.ribaf.2020.101235.
Biais, B., Foucault, T., Moinas, S., 2015. Equilibrium fast trading. Comput. Manag. Sci. 116, 292–313. http://dx.doi.org/10.1016/J.JFINECO.2015.03.004.

Blundell, R., Bond, S., 1998. Initial conditions and moment restrictions in dynamic panel data models. J. Econometrics 87, 115–143. http://dx.doi.org/10.1016/S0304-4076(98)00009-8.

Brogaard, J.A., 2010. High Frequency Trading and Its Impact on Market Quality. Working Paper, Northwestern University, Kellogg School of Management.

Brogaard, J., Hendershott, T., Riordan, R., 2014. High-frequency trading and price discovery. Rev. Financ. Stud. 27, 2267–2306. http://dx.doi.org/10.1093/RFS/HHU032.

Carrion, A., 2013. Very fast money: High-frequency trading on the NASDAQ. J. Financial Mark. 16, 680–711. http://dx.doi.org/10.1016/J.FINMAR.2013.06.005.

Cartea, A., Payne, R., Penalva, J., Tapia, M., 2019. Ultra-fast activity and market quality. J. Bank. Financ. 99, 157–181. http://dx.doi.org/10.1016/j.jbankfin.2018.12.003.

Cartea, A., Penalva, J., 2012. Where is the value in high frequency trading? Q. J. Finance 2, 1–46. http://dx.doi.org/10.1142/S2010139212500140.

CFTC, 2012. Sub-Committee on Automated and High Frequency Trading. U.S. Commodity Futures Trading Commission, URL: https://www.cftc.gov/sites/default/files/idc/groups/public/@newsroom/documents/file/tac103012_wg1.pdf.

Chancharat, S., Phuensane, P., 2021. High frequency trading and market quality in the thai stock exchange. GMSARN Int. J. 15, 244–249.

Conrad, J., Wahal, S., Xiang, J., 2015. High-frequency quoting, trading, and the efficiency of prices. J. Financ. Econ. 116 (2), 271–291. http://dx.doi.org/10.1016/j.jfineco.2015.02.008.

Desagre, C., D'Hondt, C.D., Petitjean, M., 2022. The rise of fast trading: Curse or blessing for liquidity? Finance http://dx.doi.org/10.3917/fina.pr.i.

Ding, M., Suardi, S., Xu, C., Zhang, D., 2021. Large-caps liquidity provision, market liquidity and high-frequency market makers' trading behaviour. Eur. J. Finance http://dx.doi.org/10.1080/1351847X.2021.1988670.

Dubey, R.K., Babu, A.S., Jha, R.R., Varma, U., 2021. Algorithmic trading efficiency and its impact on market-quality. Asia-Pac. Financ. Mark. http://dx.doi.org/10.1007/s10690-021-09353-5.

Dubey, R.K., Chauhan, Y., Syamala, S.R., 2017. Evidence of algorithmic trading from Indian equity market: Interpreting the transaction velocity element of financialization. Res. Int. Bus. Finance 42, 31–38. http://dx.doi.org/10.1016/j.ribaf.2017.05.014.

Egginton, J.F., Van Ness, B.F., Van Ness, R.A., 2016. Quote stuffing. Financ. Manage. 45, 583–608. http://dx.doi.org/10.1111/FIMA.12126.

Ekinci, C., Ersan, O., 2018. A new approach for detecting high-frequency trading from order and trade data. Finance Res. Lett. 24, 313–320. http://dx.doi.org/10.1016/j.frl.2017.09.020.

Ekinci, C., Ersan, O., 2022. High-frequency trading and market quality: The case of a "slightly exposed" market. Int. Rev. Financ. Anal. 79, 102004. http://dx.doi.org/10.1016/j.irfa.2021.102004.

ESMA, 2021. MiFID II/MiFIR Review Report on Algorithmic Trading. European Securities and Markets Authority.

Frino, A., Mollica, V., Webb, R., 2014. The impact of co-location of securities exchanges' and traders' computer servers on market liquidity. J. Futures Mark. 34, 20–33. http://dx.doi.org/10.1002/FUT.21631.

Goldstein, M., Kwan, A., Philip, R., 2017. High-frequency trading strategies. SSRN Electron. J. http://dx.doi.org/10.2139/ssrn.2973019.

Gomber, P., Arndt, B., Lutat, M., Uhle, T., 2011. High-frequency trading. SSRN Electron. J. http://dx.doi.org/10.2139/ssrn.1858626.

Hagströmer, B., Nordén, L., 2013. The diversity of high-frequency traders. J. Financial Mark. 16, 741–770. http://dx.doi.org/10.1016/J.FINMAR.2013.05.009.

Hasbrouck, J., 2018. High-frequency quoting: Short-term volatility in bids and offers. J. Financ. Quant. Anal. 53 (2), 613–641. http://dx.doi.org/10.1017/S0022109017001053.

Hasbrouck, J., Saar, G., 2013. Low-latency trading. J. Financial Mark. 16, 646–679. http://dx.doi.org/10.1016/j.finmar.2013.05.003.

Hendershott, T., Jones, C.M., Menkveld, A.J., 2011. Does algorithmic trading improve liquidity? J. Finance 66, 1–33. http://dx.doi.org/10.1111/j.1540-6261.2010.01624.x.

Hendershott, T., Riordan, R., 2013. Algorithmic trading and the market for liquidity. J. Financ. Quant. Anal. 48, 1001–1024. http://dx.doi.org/10.1017/S0022109013000471.

Hoffmann, P., 2014. A dynamic limit order market with fast and slow traders. J. Financ. Econ. 113, 156–169. http://dx.doi.org/10.1016/J.JFINECO.2014.04.002.

Hruška, J., 2016. Dynamics of liquidity on German stock market under the influence of HFT. Procedia – Soc. Behav. Sci. 220, 134–141. http://dx.doi.org/10.1016/j.sbspro.2016.05.477.

Jain, A., Jain, C., Khanapure, R.B., 2021. Do algorithmic traders improve liquidity when information asymmetry is high? Q. J. Finance 11, 2050015. http://dx.doi.org/10.1142/S2010139220500159.

Jarnecic, E., Snape, M., 2014. The provision of liquidity by high-frequency participants. Financ. Rev. 49, 371–394. http://dx.doi.org/10.1111/fire.12040.

Jovanovic, B., Menkveld, A., 2016. Middlemen in limit-order markets. SSRN Electron. J. http://dx.doi.org/10.2139/ssrn.1624329.

Kirilenko, A., Kyle, A.S., Samadi, M., Tuzun, T., 2017. The flash crash: High-frequency trading in an electronic market. J. Finance 72, 967–998. http://dx.doi.org/10.1111/jofi.12498.

Kwan, A., Philip, R., 2015. High-Frequency Trading and Execution Costs. Working Paper, University of Sydney.

Ladley, D., 2020. The high frequency trade off between speed and sophistication. J. Econom. Dynam. Control 116, 103912. http://dx.doi.org/10.1016/j.jedc.2020.103912.

Le, A.T., Le, T.-H., Liu, W.-M., Fong, W.L.K.Y., 2021. Dynamic limit order placement activities and their effects on stock market quality. Ann. Oper. Res. http://dx.doi.org/10.1007/s10479-021-04282-y.

Lewis, M., 2014. Flash Boys: A Wall Street Revolt. W. W. Norton & Company, New York, http://dx.doi.org/10.5860/choice.52-0941.

Li, K., 2021. Does high-frequency trading impede order execution in the stock market? Procedia Comput. Sci. 187, 501–506. http://dx.doi.org/10.1016/j.procs.2021.04.090.

Lyle, M.R., Naughton, J.P., Weller, B.M., 2015. How does algorithmic trading improve market quality? SSRN Electron. J. http://dx.doi.org/10.2139/ssrn.2587730.

SEC, 2010. Concept Release on Equity Market Structure. Release No. 34-61358; File No. S7-02-10. Securities and Exchange Commission.

Van Ness, B., Van Ness, R., Watson, E.D., 2015. Canceling liquidity. J. Financ. Res. 38, 3–33. http://dx.doi.org/10.1111/jfir.12051.